NATIONAL BOARD OF ACCREDITATION

Data Capturing Points of the Program Applied for NBA Accreditation– Tier I/II UG (Engineering) Institute Programs

Program Name : ELECTRONICS AND COMMUNICATION ENGINEERING	**Discipline**: Engineering & Technology
Level : Under Graduate	**Tier**: 1
Application No: 11860	**Date of Submission**: 20-05-2026

PART A- Profile of the Institute

A1.Name of the Institute: KLE Technological University	
Year of Establishment : 1947	Location of the Institute: BVB Campus, Vidyanagar, Hubballi
A2. Institute Address:KLE Technological University B.V. Bhoomaraddi College Campus , Vidyanagar Hubballi -580031	
City:Dharwad	State:Karnataka
Pin Code:580031	Website:www.kletech.ac.in
Email:vc@kletech.ac.in	Phone No(with STD Code):0836-2378102
A3. Name and Address of the Affiliating University (if any):	
Name of the University :	City:
State :	Pin Code: 0
A4. Type of the Institution: University	
A5. Ownership Status: Self financing	

A6. Details of all Programs being Offered by the Institution:

- No. of UG programs: **11**
- No. of PG programs: **8**

Table No. A6.1: List of all programs offered by the Institute.

Sr.No.	Discipline	Level of program	Name of the program	Year of Start	Year of Closed	Name of The Department
1	Architecture	UG	Architecture	2015	--	Architecture
2	Computer Application	PG	Master of Computer Application	2015	--	Computer Application
3	Computer Application	PG	Master of Computer Applications (Integrated)	2024	--	Computer Application
4	Engineering & Technology	PG	Advanced Manufacturing Systems	2020	--	Mechanical Engineering
5	Engineering & Technology	UG	Automation & Robotics	2015	--	Automation and Robotics
6	Engineering & Technology	UG	Biotechnology	2015	--	Biotechnology
7	Engineering & Technology	UG	Civil Engineering	2015	--	Civil Engineering
8	Engineering & Technology	UG	Computer Science and Engineering	2015	--	Computer Science and Engineering
9	Engineering & Technology	PG	Computer Science and Engineering	2015	--	Computer Science and Engineering

10	Engineering & Technology	UG	Computer Science and Engineering (Artificial Intelligence)	2021	--	Computer Science and Engineering (Artificial Intelligence)
11	Engineering & Technology	PG	Design Engineering	2020	--	Mechanical Engineering
12	Engineering & Technology	UG	Electrical & Electronics Engineering	2015	--	Electrical and Electronics Engineering
13	Engineering & Technology	UG	ELECTRONICS AND COMMUNICATION ENGINEERING	2015	--	Electronics and Communication Engineering
14	Engineering & Technology	UG	Electronics and Communication Engineering (Industry Integrated)	2020	--	Electronics and Communication Engineering
15	Engineering & Technology	UG	Electronics Engineering (VLSI Design and Technology)	2022	--	Electronics Engineering (VLSI Design and Technology)
16	Engineering & Technology	UG	Mechanical Engineering	2015	--	Mechanical Engineering
17	Engineering & Technology	PG	Structural Engineering	2015	--	Civil Engineering
18	Engineering & Technology	PG	VLSI Design & Embedded Systems	2015	--	Electronics and Communication Engineering
19	Management	PG	Master of Business Administration	2015	--	Management

A7. Programs to be considered for Accreditation vide this Application:

Table No. A7.1: List of programs to be considered for accreditation.

Name of the Department	**Having Allied Departments**	**Name of the Program**	**Program Level**
Biotechnology	No	Biotechnology	UG
Electronics and Communication Engineering	Yes	ELECTRONICS AND COMMUNICATION ENGINEERING	UG
Computer Science and Engineering	Yes	Computer Science and Engineering	UG

Table No. A7.2: Allied Department(s) to the Department of the program considered for accreditation as above.
Cluster ID. Name of the Department (in table no. A7.1) Name of allied Departments/Cluster (for table no. A7.1)

Allied Department/Cluster Name	**Program Name**	**Program Level**
Electronics Engineering (VLSI Design and Technology)	Electronics Engineering (VLSI Design and Technology)	UG

PART-B: Program information

B1. Provide the Required Information for the Program Applied For:

Table No. B1: Program details.
A. List of the Programs Offered by the Department:

SR.NO.	PROGRAM NAME	PROGRAM APPLIED LEVEL	YEAR OF START / YEAR OF CLOSED	SANCTIONED INTAKE	INCREASE/DECREASE INTAKE (if any)	YEAR OF INCREASE/DECREASE	CURRENT INTAKE	YEAR OF AICTE APPROVAL	AICTE/COMPETENT AUTHORITY ARROVAL DETAILS	ACCREDITATION STATUS	FROM	TO	NO. OF TIMES PROGRAM ACCREDITED	PRC DUR
1	ELECTRONICS AND COMMUNICATION ENGINEERING	UG	2015 / --	240	Yes	2025	540	2025	F.No. South-West/1-44638523851/2025/EOA 21-Mar-2025	Granted accreditation for 3 years for the period (specify period)	2023	2026	2	4

Sanctioned Intake for Last Five Years for the VLSI Design & Embedded Systems	
Academic Year	**Sanctioned Intake**
2025-26	540
2024-25	360
2023-24	360
2022-23	300
2021-22	300
2020-21	300

List of the Allied Departments/Cluster and Programs:

SR.NO.	ALLIED DEPARTMENT NAME	PROGRAM NAME	PROGRAM APPLIED LEVEL	YEAR OF START / YEAR OF CLOSED	SANCTIONED INTAKE	INCREASE/DECREASE INTAKE (if any)	YEAR OF INCREASE/DECREASE	CURRENT INTAKE	YEAR OF AICTE APPROVAL	AICTE/COMPETENT AUTHORITY ARROVAL DETAILS	ACCREDITATION STATUS	FROM	TO	NO. OF TIMES PROGRAM ACCREDITE
1	Electronics Engineering (VLSI Design and Technology)	Electronics Engineering (VLSI Design and Technology)	UG	2022 / --	60	No	NA	60	2022	South-West/1-10970137717/2022/EOA	Not eligible for accreditation	--	--	0

B2. Detail of Head of the Department for the program under consideration:

A. Name of the HoD :	Dr. Suneeta V. Budihal
B. Nature of appointment:	Regular
C. Qualification:	M.E. and Ph.D.

B3. Program Details

Table No.B3.1: Admission details for the program excluding those admitted through multiple entry and exit points.

Item (Information to be provided cumulatively for all the shifts with explicit headings, wherever applicable)	2025-26 (CAY)	2024-25 (CAYm1)	2023-24 (CAYm2)	2022-23 (CAYm3)	2021-22 (CAYm4)	2020-21 (CAYm5)	2019-20 (CAYm6)

N=Sanctioned intake of the program (as per AICTE /Competent authority)	540	360	360	300	300	300	300
N1=Total no. of students admitted in the 1st year minus the no. of students, who migrated to other programs/ institutions plus no. of students, who migrated to this program	506	351	360	301	299	299	289
N2=Number of students admitted in 2nd year in the same batch via lateral entry including leftover seats	0	40	30	35	36	16	27
N3=Separate division if any	0	0	1	0	0	19	0
N4=Total no. of students admitted in the 1st year via all supernumerary quotas	0	0	0	0	0	0	0
Total number of students admitted in the program (N1 + N2 + N3 + N4) - excluding those admitted through multiple entry and exit points.	506	391	391	336	335	334	316

CAY= Current Academic Year. CAYm1= Current Academic Year Minus 1 CAYm2= Current Academic Year Minus 2. LYG= Last Year Graduate. LYGm1= Last Year Graduate Minus 1. LYGm2= Last Year Graduate Minus 2.

B4. Enrolment Ratio in the First Year

Table No. B4.1: Student enrolment ratio in the 1st year.

Year of entry	**N (From Table 4.1)**	**N1 (From Table 4.1)**	**N4 (From Table 4.1)**	**Enrollment Ratio [(N1/N)*100]**
2025-26 (CAY)	540	506	0	93.70
2024-25 (CAYm1)	360	351	0	97.50
2023-24 (CAYm2)	360	360	0	100.00

Average [(ER1 + ER2 + ER3) / 3] = 97.07≡ 20.00

B5. Success Rate of the Students in the Stipulated Period of the Program

Table No.B5.1: The success rate in the stipulated period of a program.

Item	**(2021-22) LYG**	**(2020-21) LYGm1**	**(2019-20) LYGm2**
A*= (No. of students admitted in the 1st year of that batch and those actually admitted in the 2nd year via lateral entry, plus the number of students admitted through multiple entry (if any) and separate division if applicable, minus the number of students who exited through multiple entry (if any).	336.00	316.00	327.00
B=No. of students who graduated from the program in the stipulated course duration	303.00	281.00	293.00
Success Rate (SR)= (B/A) * 100	90.18	88.92	89.60

Average SR of three batches ((SR_1+ SR_2+ SR_3)/3): 89.57

B6. Academic Performance of the First-Year Students of the Program

Table No.B6.1: Academic Performance of the First-Year Students of the Program.

Academic Performance	**CAYm1(2024-25)**	**CAYm2(2023-24)**	**CAYm3 (2022-23)**
Mean of CGPA or mean percentage of all successful students(X)	7.81	7.83	7.96
Y=Total no. of successful students	342.00	351.00	296.00
Z=Total no. of students appeared in the examination	351.00	360.00	301.00
API [X*(Y/Z)]	7.61	7.63	7.83

Average API[(AP1+AP2+AP3)/3] : 7.69

B7: Academic Performance of the Second Year Students of the Program

Table No.B7.1: Academic Performance of the Second Year Students of the Program.

Academic Performance	CAYm1 (2024-25)	CAYm2 (2023-24)	CAYm3 (2022-23)
X=(Mean of 2nd year grade point average of all successful students on a 10-point scale) or (Mean of the percentage of marks of all successful students in 2rd year/10)	7.93	7.83	7.58
Y=Total no. of successful students	363.00	316.00	314.00
Z=Total no. of students appeared in the examination	381.00	330.00	318.00
API [X * (Y/Z)]	7.56	7.50	7.48

Average API [(AP1 + AP2 + AP3)/3] : 7.51

B8. Academic Performance of the Third Year Students of the Program

Table No.B8.1: Academic Performance of the Third Year Students of the Program

Academic Performance	CAYm1 (2024-25)	CAYm2 (2023-24)	CAYm3 (2022-23)
X=(Mean of 3rd year grade point average of all successful students on a 10-point scale) or (Mean of the percentage of marks of all successful students in 3rd year/10)	8.06	7.96	7.91
Y=Total no. of successful students	312.00	311.00	290.00
Z=Total no. of students appeared in the examination	316.00	314.00	294.00
API [X*(Y/Z)]:	7.96	7.88	7.80

Average API [(AP1 + AP2 + AP3)/3] : 7.88

B9. Placement, Higher Studies, and Entrepreneurship

Table No.B9.1: Placement, higher studies, and entrepreneurship details.

Item	LYG (2021-22)	LYGm1(2020-21)	LYGm2(2019-20)
FS*=Total no. of final year students	336.00	316.00	327.00
X=No. of students placed	179.00	188.00	219.00
Y=No. of students admitted to higher studies	7.00	9.00	4.00
Z= No. of students taking up entrepreneurship	0.00	0.00	0.00
Placement Index(P) = (((X + Y + Z)/FS) * 100):	55.36	62.34	68.20

Average Placement Index = (P_1 + P_2 + P_3)/3: 61.97 Placement Index Points:

PART C: Faculty Details in Department and Allied Departments

(Data to be filled in for the Department and Allied Departments)

C1. Faculty details of Department and Allied Departments

Table No.C1: Faculty details in the Department for the past 3 years including CAY

Sr.No	Name of the Faculty	PAN No.	Highest degree	University	Area of Specialization	Date of Joining in this Institution	Experience in years in current institute	Designation at Time Joining in this Institution	Present Designation	The date on which Designated as Professor/ Associate Professor if any	Nature of Association (Regular/ Contract/ Ad hoc)	Currently Associated (Y/N)	In case of NO, Date of Leaving	IS HOD?

1	Dr. Suneeta V. Budihal	XXXXXXX02B	M.E. and Ph.D.	VTU	Wireless Communication	07/09/2002	23.8	Lecturer	Professor	01/11/2018	Regular	Yes		Yes
2	Dr. R. M. Banakar	XXXXXXX98M	M.Tech and Ph.D.	IIT-DELHI	VLSI and Communication systems	07/09/1988	37.8	Lecturer	Professor	02/08/2004	Contractual Fulltime	Yes		No
3	Dr. Nalini C. Iyer	XXXXXXX64B	M.Tech and Ph.D.	VTU	Network Security	12/09/1988	36.11	Lecturer	Professor	21/04/2014	Regular	No	29/08/2025	No
4	Dr. Uma K. Mudenagudi	XXXXXXX38F	M.Tech and Ph.D.	IIT-Delhi	Computer Vision	20/11/1989	36.5	Lecturer	Professor	25/02/2008	Regular	Yes		No
5	Dr. A.V. Nandi	XXXXXXX68R	M.Tech and Ph.D.	VTU	Low Power VLSI	08/10/1992	33.7	Lecturer	Professor	26/11/2010	Regular	Yes		No
6	Dr. P. C. Nissimagoudar	XXXXXXX95A	M.E. and Ph.D.	VTU	Embedded Systems	23/09/1997	28.7	Lecturer	Professor	01/03/2023	Regular	Yes		No
7	Dr. Ujwala Patil	XXXXXXX07K	M.Tech and Ph.D.	VTU	Image Processing	10/03/2003	23.2	Lecturer	Professor	01/11/2018	Regular	Yes		No
8	Dr. S. R. Nirmala	XXXXXXX48E	M.Tech and Ph.D.	IIT-Guwahati	Speech processing	02/07/2018	7.10	Associate Professor	Professor	01/03/2023	Regular	Yes		No
9	Dr. Sujata N. Patil	XXXXXXX85A	M.Tech and Ph.D.	KAHER	Medical Image Computing	30/08/2024	1.8	Professor	Professor	30/08/2024	Regular	Yes		No
10	Dr. Shivashankar A. Huddar	XXXXXXX63D	M.Tech and Ph.D.	VTU	MEMS	07/08/2012	13.9	Assistant Professor	Associate Professor	01/03/2023	Regular	Yes		No
11	Dr. Basawaraj	XXXXXXX58R	M.Tech and Ph.D.	Ohio State University USA	Neural Networks	09/04/2021	5.1	Assistant Professor	Associate Professor	01/03/2023	Regular	Yes		No
12	Dr. Ramakrishna Joshi	XXXXXXX07A	M.Tech and Ph.D.	VTU	5G Signal Processing	13/03/2023	3.2	Associate Professor	Associate Professor	13/03/2023	Regular	Yes		No
13	Dr. Tejas Rayanagoudar	XXXXXXX39A	M.Tech and Ph.D.	VTU	Computer vision and AI ML	20/05/2024	1.11	Associate Professor	Associate Professor	20/05/2024	Regular	Yes		No
14	Ms. R. V. Hanagal	XXXXXXX62R	M.Tech	VTU	Biomedical Instrumentation	16/07/1990	35.10	Lecturer	Assistant Professor		Contractual Fulltime	Yes		No
15	Mr. Kiran M. R.	XXXXXXX61F	M.Tech	VTU	Digital Electronics	14/01/2008	18.4	Lecturer	Assistant Professor		Regular	Yes		No

16	Mr. Shamshuddin K.	XXXXXXX28P	M.Tech	VTU	Computer Networking Engineering	20/07/2010	15.9	Lecturer	Assistant Professor		Regular	Yes		No
17	Ms. Bhagyashree Kinnal	XXXXXXX02C	M.Tech	VTU	VDES	02/08/2010	15.9	Lecturer	Assistant Professor		Regular	Yes		No
18	Mr. Kaushik Mallibhat	XXXXXXX48M	M.Tech	VTU	VDES	07/05/2012	14	Assistant Professor	Assistant Professor		Regular	Yes		No
19	Ms. Preeti S. Pillai	XXXXXXX37N	M.Tech	VTU	VDES	21/09/2012	13.7	Lecturer	Assistant Professor		Regular	Yes		No
20	Ms. Shraddha B. Hiremath	XXXXXXX46K	M.Tech	VTU	VDES	01/08/2012	13.9	Lecturer	Assistant Professor		Regular	Yes		No
21	Ms. Nikita K. Patil	XXXXXXX90R	M.Tech	VTU	VDES	01/08/2015	10.9	Lecturer	Assistant Professor		Regular	Yes		No
22	Mr. Satish S. Chikkamath	XXXXXXX73D	M.Tech	VTU	Industrial Automation and Robotics	22/07/2014	11.9	Lecturer	Assistant Professor		Regular	Yes		No
23	Mr. Gireesha H. M.	XXXXXXX25C	M.Tech	VTU	Biomedical Signal Processing and Instrumentation	01/08/2015	10.9	Assistant Professor	Assistant Professor		Regular	Yes		No
24	Ms. Jyoti Ravikumar	XXXXXXX30G	M.Tech	VTU	VDES	04/08/2014	11.9	Assistant Professor	Assistant Professor		Regular	Yes		No
25	Ms. Rajeshwari Mattimani	XXXXXXX29N	M.Tech	VTU	VDES	04/08/2014	11.9	Assistant Professor	Assistant Professor		Regular	Yes		No
26	Mr. Ramesh A. Tabib	XXXXXXX91M	M.Tech	VTU	Digital Electronics	01/06/2017	8.11	Assistant Professor	Assistant Professor		Regular	Yes		No
27	Ms.Supriya Katwe	XXXXXXX65Q	M.Tech	VTU	Digital Electronics	01/10/2020	5.7	Assistant Professor	Assistant Professor		Regular	Yes		No
28	Mr. Shivaprasad Chennagi	XXXXXXX93B	M.Tech	VTU	Micro Electronics and control Systems	10/04/2021	5.1	Assistant Professor	Assistant Professor		Regular	Yes		No
29	Ms. Rajeshwari Karnam	XXXXXXX48N	M.Tech	VTU	Digital Electronics	02/11/2022	3.6	Assistant Professor	Assistant Professor		Regular	Yes		No
30	Ms. Nivedita Shettar	XXXXXXX38R	M.Tech	VTU	VDT	04/12/2023	2.5	Assistant Professor	Assistant Professor		Regular	Yes		No
31	Ms. Saiarpita Rathod	XXXXXXX91A	M.Tech	VTU	VDES	10/11/2023	2.6	Assistant Professor	Assistant Professor		Regular	Yes		No
32	Ms. Deepa Betageri	XXXXXXX84M	M.Tech	KLETU	Digital Electroonics	01/01/2024	2.4	Assistant Professor	Assistant Professor		Regular	Yes		No
33	Mr. Mohammed Azharuddin	XXXXXXX39K	M.Tech	VTU	Computer science Engineering	18/10/2023	2.6	Assistant Professor	Assistant Professor		Regular	Yes		No

34	Ms. Sushma Garawad	XXXXXXX54R	M.Tech	KLETU	VDES	01/11/2023	2.6	Assistant Professor	Assistant Professor		Regular	Yes		No
35	Ms. Lydia R. Darla	XXXXXXX52G	M.Tech	KLETU	VDES	07/11/2023	2.6	Assistant Professor	Assistant Professor		Regular	Yes		No
36	Mr. Sachidanand B. N.	XXXXXXX68J	M.Tech	VTU	Digital Electronics	31/08/2024	1.8	Assistant Professor	Assistant Professor		Regular	Yes		No
37	Ms. Chaitra Anantpur	XXXXXXX15K	M.Tech	VTU	Digital Communication	04/11/2024	1.6	Assistant Professor	Assistant Professor		Regular	Yes		No
38	Mr. Amit Telgar	XXXXXXX33P	M.Tech	VTU	Digital Communication	08/10/2024	1.7	Assistant Professor	Assistant Professor		Regular	Yes		No
39	Mr. Mani C.	XXXXXXX56K	M.Tech	VTU	Power Electronics	01/01/2025	1.4	Assistant Professor	Assistant Professor		Regular	Yes		No
40	Mr. Raghuraj Adi	XXXXXXX16P	M.Tech	VTU	VDES	06/03/2025	1.2	Assistant Professor	Assistant Professor		Regular	Yes		No
41	Ms. Neena Budiyal	XXXXXXX01G	M.Tech	KLETU	Digital Electronics	17/02/2025	1.2	Assistant Professor	Assistant Professor		Contractual Fulltime	Yes		No
42	Ms. Sujata Ayyappanavar	XXXXXXX36L	M.Tech	VTU	Digital Electronics	05/10/2024	1.7	Assistant Professor	Assistant Professor		Regular	Yes		No
43	Ms. Rohini Bhatkoorse	XXXXXXX52D	M.Tech	VTU	Communication Systems	17/03/2025	1.1	Assistant Professor	Assistant Professor		Regular	Yes		No
44	Mr. K S Shubham	XXXXXXX31F	M.Tech	KLETU	VDES	17/06/2025	0.10	Assistant Professor	Assistant Professor		Regular	Yes		No
45	Sangeeta Muttagi	XXXXXXX15J	M.Tech	VTU	Digital Electronics	19/08/2025	0.8	Assistant Professor	Assistant Professor		Regular	Yes		No
46	Preeti Jigalur	XXXXXXX55P	M.Tech	VTU	Information Technology	01/08/2025	0.9	Assistant Professor	Assistant Professor		Regular	Yes		No
47	Vinodkumar Kengeri	XXXXXXX12N	M.Tech	VTU	Computer Networks	04/08/2025	0.9	Assistant Professor	Assistant Professor		Regular	Yes		No
48	Gayatribai K	XXXXXXX50Q	M.Tech	VTU	Digital Electronics	22/08/2025	0.8	Assistant Professor	Assistant Professor		Regular	Yes		No
49	Uday Wali	XXXXXXX29N	M.Tech and Ph.D.	IIT Kharagpur	VLSI	01/04/2022	4.1	Professor	Professor	01/04/2022	Contractual Fulltime	Yes		No
50	Dr. G Priyatam Kumar	XXXXXXX77K	M.Tech and Ph.D.	VTU	Wireless Communication	20/12/1989	35.10	Lecturer	Professor	30/06/2011	Regular	No	31/10/2025	No
51	Dr. R. B. Shettar	XXXXXXX49A	M.Tech and Ph.D.	VTU	VLSI Architecture	21/10/1991	34.2	Lecturer	Professor	24/04/2014	Regular	No	18/12/2025	No
52	Dr. Rohini S. Hongal	XXXXXXX11N	M.Tech and Ph.D.	VTU	Embedded Security	13/02/2003	22.10	Lecturer	Professor	06/04/2021	Regular	No	18/12/2025	No

53	Ms. Poornima Patil	XXXXXXX96D	M.Tech	VTU	VDES	07/08/2023	2	Assistant Professor	Assistant Professor		Regular	No	21/08/2025	No
54	Ms. Sumita Guddin	XXXXXXX23N	M.Tech	VTU	Computer Science	03/10/2023	1.9	Assistant Professor	Assistant Professor		Regular	No	09/07/2025	No
55	Dr. Tanuja R. Patil	XXXXXXX50M	M.Tech and Ph.D.	VTU	Image Processing	16/03/1994	32.2	Lecturer	Associate Professor	01/11/2011	Regular	Yes		No
56	Ms. Vidyashri Math	XXXXXXX86D	M.Tech	VTU	Information Technology	09/04/2021	5.1	Assistant Professor	Assistant Professor		Regular	Yes		No
57	Ms. Sheela A Badagi	XXXXXXX37R	M.Tech	VTU	Computer Science Engg.	03/09/2019	6.8	Assistant Professor	Assistant Professor		Regular	Yes		No
58	Ms. Nirmala Koliwad	XXXXXXX09R	M.Tech	VTU	Computer Science	19/12/2022	3.4	Assistant Professor	Assistant Professor		Regular	Yes		No
59	Ms. Heena Shirahatti	XXXXXXX69C	M.Tech	VTU	Digital Electronics	01/07/2022	3.10	Assistant Professor	Assistant Professor		Regular	Yes		No
60	Supriya P	XXXXXXX90H	M.Tech	VTU	Digital Communication	30/08/2025	0.8	Assistant Professor	Assistant Professor		Regular	Yes		No
61	Renuka Bariker	XXXXXXX58A	M.Tech	VTU	Digital Communication	30/08/2025	0.8	Assistant Professor	Assistant Professor		Regular	Yes		No
62	Ms.Padmaja Kallimani	XXXXXXX83J	M.Tech	VTU	Power Systems Engineering	01/08/2018	6.10	Assistant Professor	Assistant Professor		Regular	No	31/05/2025	No
63	Ms.Pooja Chandaragi	XXXXXXX04F	M.Tech	KLETU	VDES	20/06/2022	2.11	Assistant Professor	Assistant Professor		Regular	No	31/05/2025	No
64	Mr. Sangmesh V Melinmani	XXXXXXX65A	M.Tech	VTU	VLSI Design and Testing	20/06/2022	2.11	Assistant Professor	Assistant Professor		Regular	No	31/05/2025	No
65	Ms. Jayashree Mallidu	XXXXXXX65E	M.Tech	VTU	Digital Electronics	01/08/2018	6.10	Assistant Professor	Assistant Professor		Regular	No	31/05/2025	No
66	Mr. Akash Kulkarni	XXXXXXX74A	M.Tech	VTU	RF & Microwave Communication	01/10/2019	4.9	Assistant Professor	Assistant Professor		Regular	No	27/07/2024	No
67	Dr. Hemantraj N. Kelagadi	XXXXXXX09K	M.Tech and Ph.D.	VTU	Wireless Sensor Networks	08/10/2007	16.9	Assistant Professor	Assistant Professor		Regular	No	31/07/2024	No
68	Ms. Sahana E Punagin	XXXXXXX90K	M.Tech	KLETU	Digital Electronics	09/04/2021	3.4	Assistant Professor	Assistant Professor		Regular	No	08/08/2024	No
69	Mr. Rohit Kalyani	XXXXXXX73M	M.Tech and Ph.D.	VTU	Control Systems	01/06/2018	6.7	Assistant Professor	Assistant Professor		Regular	No	31/12/2024	No
70	Mr. Shivaraj Hublikar	XXXXXXX58H	M.Tech	VTU	Microelectronics Engineering	29/08/2002	22.5	Assistant Professor	Assistant Professor		Regular	No	31/01/2025	No
71	Mr. R. M. Shet	XXXXXXX75N	M.Tech and Ph.D.	VTU	Control Systems	25/08/2004	20.5	Lecturer	Assistant Professor		Regular	No	31/01/2025	No

72	Dr. Saroja Siddamal	XXXXXXX40F	M.Tech and Ph.D.	JNTU	VLSI architecture & Signal Processing	04/10/2002	21.9	Lecturer	Professor	27/06/2016	Regular	No	01/07/2024	No
73	Dr. Vijay H. M.	XXXXXXX32B	M.Tech and Ph.D.	VIT Vellore	Semiconductor memories	10/04/2021	3.4	Assistant Professor	Associate Professor	01/03/2023	Regular	No	30/08/2024	No
74	Mr. Shashidhar N.	XXXXXXX36B	M.Tech	VTU	Digital Electronics	01/09/2017	7	Assistant Professor	Assistant Professor		Regular	No	30/08/2024	No
75	Dr. Nalini C. Iyer	XXXXXXX64B	M.Tech and Ph.D.	VTU	Network Security	30/08/2025	0.8	Professor	Professor	30/08/2025	Contractual Fulltime	Yes		No
76	Mrs. Renuka Ganiger	XXXXXXX08Q	M.Tech	VTU	Digital Electronics & Communication Systems	01/09/2022	2	Assistant Professor	Assistant Professor		Regular	No	02/09/2024	No
77	Mrs Shashikala V. Budni	XXXXXXX54G	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Science and Engineering	30/07/2007	18.9	Lecturer	Assistant Professor		Regular	Yes		No
78	Mrs Sunita K. Salimath	XXXXXXX54J	M.Tech	Visvesvaraya Technological University, Belagavi	Computer Network Engineering	15/04/1998	28.1	Lecturer	Assistant Professor		Regular	Yes		No
79	Ms. Geeta Sannakki	XXXXXXX29K	M.Tech	Visvesvaraya Technological University, Belagavi	Electronics and Communication Engineering	29/07/2024	1.9	Assistant Professor	Assistant Professor		Regular	Yes		No
80	Ms. Namrata Hiremath	XXXXXXX71N	M.Tech	Visvesvaraya Technological University, Belagavi	Digital Electronics	01/08/2007	17.5	Lecturer	Assistant Professor		Regular	No	01/01/2025	No
81	Sujata R kulkarni	XXXXXXX72M	M.Tech	VTU	Computer Science and Engineering	16/06/2008	17.11	Lecturer	Assistant Professor		Regular	Yes		No

Table No.C2: Faculty details of Allied Departments for the past 3 years including CAY.

Sr.No	**Name of the Faculty**	**PAN No.**	**APAAR faculty ID*(if any)**	**Highest degree**	**University**	**Area of Specialization**	**Date of Joining in this Institution**	**Experience in years in current institute**	**Designation at Time Joining in this Institution**	**Present Designation**	**The date on which Designated as Professor/ Associate Professor if any**	**Nature of Association (Regular/ Contract/ Ad hoc)**	**Currently Associated (Y/N)**	**In case of NO, Date of Leaving**	**IS HOD?**
1	Dr. Sujata Kotabagi	XXXXXXX67B	NA	MS and PhD	VTU	Analog & Mixed Mode VLSI	29/08/2002	23.8	Assistant Professor	Professor	01/11/2018	Regular	Yes		Yes

2	Dr. Vijay H. M.	XXXXXXX32B	NA	M.Tech and Ph.D.	VIT Vellore	Semiconductor memories	31/08/2024	1.8	Associate Professor	Associate Professor		Regular	Yes		No
3	Ms. Asmita A.	XXXXXXX04E	NA	M.Tech	VTU	Industrial Electronics	01/02/2025	1.3	Assistant Professor	Assistant Professor		Regular	Yes		No
4	Ms. Sonia Bhusare	XXXXXXX55D	NA	M.Tech	KLETU	VLSI and Embedded System	17/02/2025	1.2	Assistant Professor	Assistant Professor		Regular	Yes		No
5	Dr. Suhas B. Shirol	XXXXXXX86A	NA	M.Tech and Ph.D.	VTU	VLSI Architecture	06/08/2012	13.4	Assistant Professor	Assistant Professor		Regular	No	30/12/2025	No
6	Mr. Shashidhar N.	XXXXXXX36B	NA	M.Tech	VTU	Digital Electronics	31/08/2024	1.3	Assistant Professor	Assistant Professor		Regular	No	30/12/2025	No

C2. Student-Faculty Ratio (SFR)

No. of UG(Engineering) programs in Department including allied departments/ clusters (UGn):
UG1=1st UG program
UGn=nth UG program
B= No. of Students in UG 2nd year (ST)
C= No. of Students in UG 3rd year (ST)
D= No. of Students in UG 4th year (ST)
No. of PG (Engineering) programs in Department including allied departments/ clusters (PGm):
PG1=1st PG program.
PGm=mth PG program
A= No. of Students in PG 1st year
B= No. of Students in PG 2nd year
Student Faculty Ratio **(SFR)** = S/F
S= No. of students of all programs in the Department including all students of allied departments/clusters.
No. of students (ST)=Sanctioned Intake (SA)+ Actual admitted students via lateral entry including leftover seats (L) if any (limited to 10 % of SA)
Students who admitted under supernumerary quotas (SNQ, EWS, etc) will not be considered in calculating SFR value. Those students are exempted.
F=Total no. of regular or contractual faculty members (Full Time) in the Department, including allied departments/clusters (excluding first year faculty (The faculty members who have a 100% teaching load in the first-year courses)).

No. of UG Programs in the Department2 No. of PG Programs in the Department1
Table No.C2.1: Student-faculty ratio.

Description	**CAY(2025-26)**	**CAYm1 (2024-25)**	**CAYm2 (2023-24)**
UG1.B	0	0	65
UG1.C	0	64	65
UG1.D	66	65	66
UG1: Electronics and Communication Engineering (Industry Integrated)	**66**	**129**	**196**
UG2.B	396	390	330
UG2.C	396	329	330
UG2.D	330	328	330
UG2: ELECTRONICS AND COMMUNICATION ENGINEERING	**1122**	**1047**	**990**
UG3.B	65	64	64

Description	CAY(2025-26)	CAYm1 (2024-25)	CAYm2 (2023-24)
UG3.C	64	65	0
UG3.D	65	0	0
UG3: Electronics Engineering (VLSI Design and Technology)	**194**	**129**	**64**
PG1.A	18	18	24
PG1.B	18	24	24
PG1: VLSI Design & Embedded Systems	**36**	**42**	**48**
DS=Total no. of students in all UG and PG programs in the Department	1224	1218	1234
AS=Total no. of students of all UG and PG programs in allied departments	194	129	64
S=Total no. of students in the Department (DS) and allied departments (AS)	**S1=** 1418	**S2=** 1347	**S3=** 1298
DF=Total no. of faculty members in the Department	60	55	55
AF= Total no. of faculty members in the allied Departments	4	4	2
F=Total no. of faculty members in the Department (DF) and allied Departments (AF)	**F1=** 64	**F2=** 59	**F3=** 57
FF=The faculty members in F who have a 100% teaching load in the first-year courses	6	4	4
Student Faculty Ratio (SFR)=S/(F-FF)	**SFR1= 24.45**	**SFR2= 24.49**	**SFR3= 24.49**
Average SFR for 3 years	**SFR= 24.48**		

C3. Faculty Qualification

- Faculty qualification index (FQI) = 2.5 * [(10X +4Y)/RF] where
- X=No. of faculty members with Ph.D. degree or equivalent as per AICTE/UGC norms.
- Y=No. of faculty members with M. Tech. or ME degree or equivalent as per AICTE/ UGC norms.
- RF=No. of required faculty in the Department including allied Departments to adhere to the 20:1 Student-Faculty ratio, with calculations based on both student numbers and faculty requirements as per section C2 of this documents: (RF=S/20).

Table No.C3.1: Faculty qualification.

Year	X	Y	RF	FQ = 2.5 x [(10X + 4Y) / RF)]
2025-26(CAY)	17	47	70.00	12.79
2024-25(CAYm1)	20	39	67.00	13.28
2023-24(CAYm2)	20	37	64.00	13.59

C4. Faculty Cadre Proportion

- Faculty Cadre Proportion is 1(RF1): 2(RF2): 6(RF3)
- RF1= No. of Professors required = 1/9 * No. of Faculty required to comply with 20:1 Student-Faculty ratio based on no. of students (S) as per C2 of this documents:.
- RF2= No. of Associate Professors required = 2/9 * No. of Faculty required to comply with 20:1 Student-Faculty ratio based on no. of students (S) as per section C2 of this documents:.
- RF3= No. of Assistant Professors required = 6/9 * No. of Faculty required to comply with 20:1 Student-Faculty ratio based on no. of students (S) as per section C2 of this documents:.
- Faculty cadre and qualification and experience should be as per AICTE/UGC norms.

Table No.C4.1: Faculty cadre proportion details.

Year	Professors		Associate Professors		Assistant Professors	
	Required RF1	Available AF1	Required RF2	Available AF1	Required RF3	Available AF3

2025-26	7.00	8.00	15.00	6.00	47.00	45.00
2024-25	7.00	12.00	14.00	6.00	44.00	38.00
2023-24	7.00	12.00	14.00	4.00	43.00	38.00
Average	RF1=7.00	AF1=10.67	RF2=14.33	AF2=5.33	RF2=44.67	AF2=40.33

C5. Visiting/Adjunct Faculty/Professor of Practice

Table No. C5.1: List of visiting/adjunct faculty/professor of practice and their teaching and practical loads.

(CAYm1)

S.No	Name of the Person	Designation	Organization	Name of the Course	No. of hours handled
1	V. D. Kulkarni	Adjunct Faculty	CoreEL Technologies PVT, Bangalore	CMOS VLSI Design, Minor Project	75.00
2	Dr.Anand Bariya	Adjunct Professor	TCS, Bangalore	CMOS ASIC Design, Project Internship	80.00
3	Vijay Gudi	Professor of Practice	Semi-Ksha Semiconductors	System Verilog and Verification, Architectural Design of Digital IC	80.00

(CAYm2)

S.No	Name of the Person	Designation	Organization	Name of the Course	No. of hours handled
1	V. D. Kulkarni	Adjunct Faculty	CoreEL Technologies PVT, Bangalore	IRP, Minor Project	80.00
2	Dr.Anand Bariya	Adjunct Professor	TCS, Bangalore	CMOS ASIC Design, Project Internship	80.00
3	Shripad Annigeri	Adjunct Professor	Micron Semiconductor India Pvt. Ltd	Project Internship	80.00

(CAYm3)

S.No	Name of the Person	Designation	Organization	Name of the Course	No. of hours handled
1	V. D. Kulkarni	Adjunct Faculty	CoreEL Technologies PVT, Bangalore	IRP, Minor Project	60.00
2	Dr.Anand Bariya	Adjunct Professor	TCS, Bangalore	Minor Project, Project internship	60.00
3	Shripad Annigeri	Adjunct Professor	Micron Semiconductor India Pvt. Ltd	Minor Project, Project internship	60.00

C6. Academic Research

Table No. C6.1: Faculty publication details.

S.No.	Item	2024-25 (CAYm1)	2023-24 (CAYm2)	2022-23 (CAYm3)
1	No. of peer reviewed journal papers published	15	13	11
2	No. of peer reviewed conference papers published	85	98	26
3	No. of books/book chapters published	8	4	1

C7. Sponsored Research Project

Table No. C7.1: List of sponsored research projects received from external agencies.

(CAYm1)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25
Ramakrishna S		School of Electronics and Communication Engineering	CoPercept-AMR: A Cooperative Perception-Based Localization Framework for Autonomous Mobile Robots in 5G-Enabled Smart Environments	Department of Telecommunication	1	3.00
						Amount received (Rs.):3.00

(CAYm2)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25
Uma Mudenagudi	Meena S M; Sujatha C	School of Electronics and Communication Engineering	KALPIT: Digital Creation to Enable Gamification Based on Indian Ethos	SERB-INAE	24	63.89
						Amount received (Rs.):63.89

(CAYm3)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25
Suneeta V Budihal	Saroja V Siddamal	School of Electronics and Communication Engineering	e Krushi Precision Agriculture	DST-SERB	36	15.68
Saroja V Siddamal	Suneeta V Budihal	School of Electronics and Communication Engineerin	MATOMAC: Development of LPWAN ASIC/SoC	MeitY	36	500.00
Ujwala Patil		School of Electronics and Communication Engineering	Research challenges and Application in Visual Intelligence	AICTE	1	3.00
						Amount received (Rs.):518.68

Total Amount (Lacs) Received for the Past 3 Years: 585.57

Note*:

- Only sponsored research projects will be considered. Infrastructure-based projects will not be considered here.

C8. Consultancy Work

Table No. C8.1: List of consultancy projects received from external agencies.

(CAYm1)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25
Ramesh A. Tabib		CEVI-SEED under SOCSE	Data and Operational Consultancy	Samsung R&D Institute India Bangalore(SRI-B)	12	28.69
						Amount received (Rs.):28.69

(CAYm2)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25
Ramesh A. Tabib		CEVI-SEED under SOCSE	Data and Operational Consultancy	Samsung R&D Institute India Bangalore(SRI-B)	12	32.13
						Amount received (Rs.):32.13

(CAYm3)

Total amount (Lacs) received for the past 3 years: 60.82

Note*:

- Only consultancy projects will be considered. Infrastructure-based projects will not be considered here.

C9. Institution Seed Money or Internal Research Grant to its Faculty for Research Work

Table No. C9.1: List of faculty members received seed money or internal research grant from the Institution.

(CAYm1)

Faculty name	Project title/ Support for Activity	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25	Amount Utilized(Lacs) i.e. 15,25,000=15.25	Outcomes of the project
Ujwala Patil	STAR Lab: Center for Intelligent Mobility (CIM)	12	28.00	11.79	Students: 12 REU; 4 MTech; 5 PhD; 8 Capstone; 140 Minor/ Major 30 research articles and 3 external project proposals
			Amount received (Rs.): 28.00		

(CAYm2)

Faculty name	Project title/ Support for Activity	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25	Amount Utilized(Lacs) i.e. 15,25,000=15.25	Outcomes of the project
Ujwala Patil	STAR Lab: Center for Intelligent Mobility (CIM)	12	52.15	4.81	Students: 14 REU; 4 MTech; 5 PhD; 2 Capstone; 130 Minor/ Major 28 research articles
			Amount received (Rs.): 52.15		

(CAYm3)

Faculty name	Project title/ Support for Activity	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25	Amount Utilized(Lacs) i.e. 15,25,000=15.25	Outcomes of the project
Ujwala Patil	STAR Lab: Center for Intelligent Mobility (CIM)	12	240.65	240.65	Students: 6 REU; 2 MTech; 5 PhD; 2 Capstone; 200 Minor/ Major 10 research articles; 6 grants applied
			Amount received (Rs.): 240.65		

Total amount (Lacs) received for the past 3 years : 320.80

PART D: Laboratory Infrastructure in the Department

(Data to be filled in for the Department)

D1. Adequate and Well-Equipped Laboratories, and Technical Manpower

Table No.D1.1: List of laboratories and technical manpower.

Sr. No	Name of the Laboratory	Number of students per set up(Batch Size)	Name of the Important Equipment	Weekly utilization status(all the courses for which the lab is utilized)	Technical Manpower Support		
					Name of the Technical staff	Designation	Qualification
1	Digital Electronics Lab	36	i. Power Supply Make ALS,5V/1.5A ii. Scientific Multiple Power supply PSD 3304 30V/2A.5V/5A,+/-15V/1A	22hrs/week (Di	Mr. Harish Hiremath	Asst. Instructor	Diploma ECE
2	Embedded System - I	72	i. Desktop Computers ii. ARM kits (LPC2148) iii. ATMEL89C51ED2	20hrs/week (Pr	Mr. Hanumanth .K	Asst. Instructor	Diploma E&C
3	Embedded System – II	72	i. Desktop Computers ii. ARM7 kits iii. ARM CORTEX M3-01 LPC1768	24hrs/week (Pr	Mr. Vinayak Arkasali	Instructor	Diploma E&C
4	VLSI Design-I	75	i. Desktop Computers ii. Cadence Tool Server	20hrs/week (Di	Mr. Mahaveer Jayakkanav	Instructor	Diploma E&C
5	VLSI Design-II	75	i. Desktop Computers ii. ALS-SDA-FPGA-04 Board Universal baseboard with Daughter Spartan 6 FPGA iii.	20hrs/week (Di	Mr. Manjunath L.K	Asst. Instructor	Diploma E&C
6	Communication & Networking	36	i. Desktop Computers ii. Juniper Switches iii. ALS SDA-XT-IOT-01 Kits iv. NI USRP-2954R	20hr/week (Jur	Mr. Shankargouda Kulkarı	Instructor	Diploma E&C
7	Analog Electronics	36	i. CRO Scientific 30MHz HM203 ii. CRO Scientific 30MHz SM203G iii. CRO Scientech 30MHz 801C iv.	28hr/week (Ana	Mr. Shriram Katti	Asst. Instructo	Diploma E&E
8	Automotive Electronics	36	HARDWARE i. Computers ii. ARMkit(LPC1768) iii. Raspberry Pi 3Board iv. Raspberry Pi 4Board v.	24hrs/week (Au	Miss. Anita	Asst. Instructo	Diploma E&C
9	Signal Processing	72	i. TMS320C6748 DSP Development Kit with XDS100 Emulator with accessories ii. Code composer 6.2.0	24hrs/week (Di	Mr. Suresh Munj	Asst. Instructor	B.Sc
10	IC Packaging	27	i. Dell OptiPlex 3020 with Intel Core i7 (4th Gen) @ 4.0 GHz, 8 GB DDR3 RAM, 500 GB storage ii. Intel Core i5	22hr/week (IC	Mr. Mandar	Asst. Instructor	Diploma E&C
11	Project Lab	35	LCR Meter (4kHz - 8MHz)	22hrs/week (Pr	Mr. Vinayak Mahale	Instructor	BE E&C

D2. Safety Measures in Laboratories

Table No. D2.1: List of various safety measures in laboratories.

Sr. No	Laboratory Name	Safety Measures

1	Digital Electronics Lab	Do’s Handle all laboratory equipment carefully. Ensure all power supplies are properly grounded. Use only regulated DC power supplies within the recommended voltage and current limits. Switch off the power supply before making or changing any circuit connections. Follow the lab manual and instructor’s instructions carefully. Maintain a logbook to record equipment usage and report faults.Check all equipment before the commencement of experiments to ensure they are in proper working condition. Report any breakage or malfunction immediately to the lab in charge. Don’ts Do not keep bags, water bottles, or food items on the laboratory tables. Do not operate equipment that is damaged or not in working condition. Do not ignore faults or breakages—always inform the lab in charge immediately.
2	Analog Electronics	Do’s Strictly follow all safety instructions given by the instructor/faculty before starting any laboratory work. Ensure the workbench is clean, dry, and free from unnecessary items. Perform experiments according to the appropriate standards and procedures. Switch OFF the power supply before connecting or modifying any circuit. Check the power supply voltage and current ratings before switching it ON. Don’ts Do not wear open-toed shoes or sandals in the laboratory. Do not handle any equipment without prior permission from the instructor/faculty. Do not ignore safety procedures or laboratory rules.
3	Communication & Networking	Do’s Keep the laboratory clean, organized, and free from cluttered cables. Ensure hands are dry before handling or operating any equipment. Switch OFF the power supply before plugging in or unplugging network hardware. Handle routers, switches, and servers carefully to prevent damage. Use strong passwords for all network configurations. Use the first aid box immediately in case of any accident or injury Don’ts Do not work with wet hands or in damp conditions. Do not drop, shake, or mishandle network devices. Do not leave cables scattered or improperly arranged in the lab
4	Signal Processing	Do’s Keep the workspace clean and free from unnecessary equipment and cables. Switch OFF the power supply before connecting or disconnecting DSP kits, CROs, or function generators. Ensure all equipment, including CROs and power supplies, are properly grounded. Don’ts Do not leave unused equipment or loose cables on the workbench. Do not connect or disconnect DSP kits, CROs, or function generators while the power supply is ON. Do not operate equipment that is not properly grounded.
5	Embedded System – I	Do’s Keep food, drinks, and bags away from the workbenches. Maintain proper lighting and ventilation in the laboratory. Always turn OFF the power supply before connecting or disconnecting wires, sensors, or boards. Handle the board with dry hands and keep all liquids away from the equipment. Ensure the USB/serial cable is connected properly before operation. Don’ts Do not touch the board’s components while it is powered ON to avoid ESD (Electrostatic Discharge) damage. Do not repeatedly plug and unplug cables, as it may cause damage to the ports or connectors. Do not operate equipment with wet hands or in the presence of liquids.
6	Embedded System – II	Do’s Keep food, drinks, and bags away from the workbenches. Maintain proper lighting and ventilation in the laboratory. Always turn OFF the power supply before connecting or disconnecting wires, sensors, or boards. Handle the board with dry hands and keep all liquids away from the equipment. Ensure the USB/serial cable is connected properly before operation. Don’ts Do not touch the board’s components while it is powered ON to avoid ESD (Electrostatic Discharge) damage. Do not repeatedly plug and unplug cables, as it may cause damage to the ports or connectors. Do not operate equipment with wet hands or in the presence of liquids.
7	VLSI Design – I	Do’s Log in to the PC using your assigned USERNAME and PASSWORD. Save all your files and work only in the space provided to you on the PC. Carry out the experiment strictly as per the given instructions. Switch OFF the machine after completing the experiment. Don’ts Do not use pen drives or external storage devices in the laboratory. Do not install any software on any machine without prior approval from the Lab In-charge or Lab Instructor. Do not delete any files or content other than your own work. Do not create your own directories on any machine. Do not change or tamper with the power connections of the machine.
8	VLSI Design – II	Do’s Log in to the PC using your assigned USERNAME and PASSWORD. Save all your files and work only in the space provided to you on the PC. Carry out the experiment strictly as per the given instructions. Switch OFF the machine after completing the experiment. Don’ts Do not use pen drives or external storage devices in the laboratory. Do not install any software on any machine without prior approval from the Lab In-charge or Lab Instructor. Do not delete any files or content other than your own work. Do not create your own directories on any machine. Do not change or tamper with the power connections of the machine.
9	Automotive Electronics	Do’s Keep the workspace clean and free from unnecessary equipment or loose cables. Ensure all equipment is properly grounded to prevent electrical shocks. Always turn OFF the power supply before connecting or disconnecting ECUs, sensors, or motor drivers. Maintain a safe distance while testing rotating sensors or actuators. Don’ts Do not expose sensors to severe vibrations or physical impact. Do not look directly into active LiDAR emitters, as it may cause eye damage. Do not handle or adjust sensors or actuators while they are powered ON.
10	IC packing lab	Do’s Keep the workspace clean and free of unnecessary tools or materials. Turn OFF the power supply before connecting or removing IC test sockets or boards. Don’ts Do not bring food, water bottles, or unrelated electronic devices near the work area. Do not connect or remove IC test sockets or boards while the power is ON.

11	Project lab	Do’s Always turn OFF the power supply before: Changing components Adjusting circuit connections Mounting or removing ICs Ensure proper grounding for instruments such as oscilloscopes and function generators. Shut down computers and instruments properly before leaving the laboratory. Report any damaged tools or faulty equipment immediately to the lab in-charge. Ensure proper ventilation when using: Etching chemicals ($FeCl_3$) PCB cleaners Isopropyl alcohol Don’ts Do not work on live circuits while changing components or ICs. Do not use damaged or faulty tools or equipment. Do not leave computers or instruments powered ON when not in use Do not use chemicals in poorly ventilated areas.

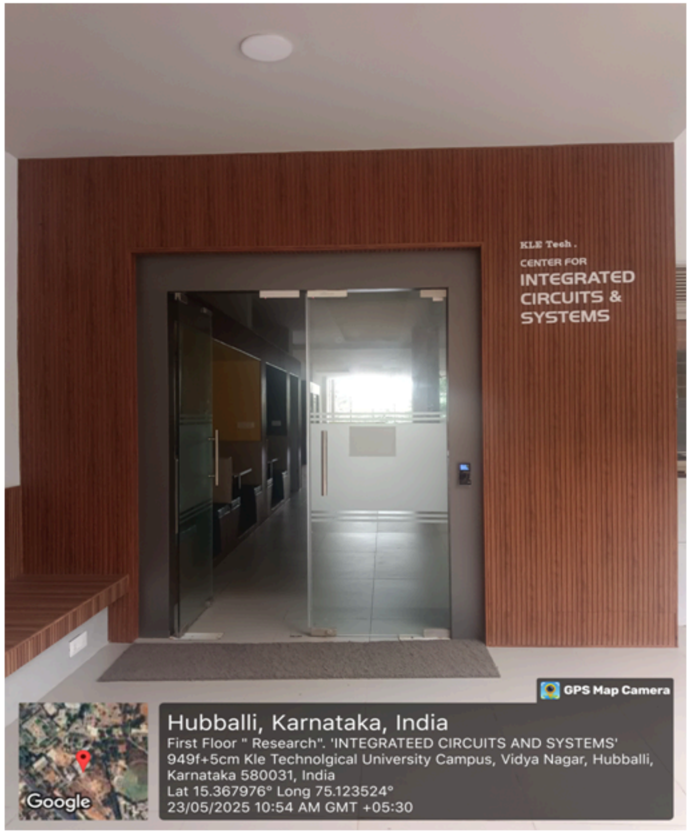
D3. Project Laboratory/Research Laboratory

Sl. No.	Name of the Lab	Area of Work	Resource	Utilization

1.	Centre for Intelligent Mobility	Perception and Control	• 3D Lidar	UG, PG & PhD Students
			• RADAR1642 Evaluation module, DCA1000EVM EVM Capture Card	
			• Real Sense Depth Camera with Operating Range (0.3m – 3m), Resolution 1280×720, USB-C Interface	
			• MTiG-710 2A8G4DK (Xsens)	
			• Xilinx PYNQ Z2	
			• Xilinx ZCU104 Evaluation kit	
			• Buggy Vehicle	
			• Logitech G29 Racing Wheel	
			• Logitech G Driving Shifter	
			• PNY Graphics Card T1000 4GB	
			• AWR Radar	
			• Q-CAR SDRS Package	
			• dSPACE-ADAS-HIL Systems	
			• Workstations(HP Z6 G4	
			• 32-bit ARM Cortex-M7 Evaluation Board	

			• Event Camera Sensor PSF138MPEV280			
			• Skateboard			
			• Autonomous Buggy			
			Connectivity		• USRPB210	UG, PG & PhD Students
		Vehicular security	• Chipwhisperer Lite-2 port with XMEGA			
		2.	Centre for Integrated Circuits and Systems	Design for Testability	• Tessent Tool	UG, PG
Signal Integrity and Electromagnetic Compatibility	• Cadence (Sigrity)			UG, PG		


KLE Tech.
CENTER FOR
INTELLIGENT MOBILITY
GPS Map Camera
Google
Hubballi, Karnataka, India
Research Lab 'CIM' 949f+5cm Kle Technolgical University
Campus, Vidya Nagar, Hubballi, Karnataka 580031, India
Lat 15.367976° Long 75.123524°
23/05/2025 10:23 AM GMT +05:30



1. **3D LiDAR:** A 3D LiDAR sensor from Velodyne is utilized for high-resolution environmental mapping and object detection. The rotating multi-beam laser architecture enables real-time acquisition of dense 3D point clouds, supporting applications in autonomous navigation, obstacle avoidance and SLAM. Its wide field of view and long-range detection capability allow accurate reconstruction of complex environments under diverse lighting conditions. The system facilitates precise localization and perception experiments by providing reliable spatial data for algorithm development and validation.

2. **RADAR AWR1642 Evaluation Module with DCA1000 EVM Capture Card:** The AWR1642 radar evaluation module is integrated with the DCA1000 EVM capture card to assess mmWave radar sensing performance for automotive and robotics applications. This setup provides real-time acquisition of raw IF radar data, enabling experiments related to range estimation, velocity profiling, and object detection. The high-frequency FMCW radar supports robust sensing in low-visibility environments, while the capture card ensures lossless high-speed data streaming for algorithm training, signal processing, and performance benchmarking.

3. **Real Sense Depth Camera (0.3 m – 3 m, 1280×720, USB-C):** An Intel RealSense depth camera is used for stereo-based depth sensing and RGB-D data acquisition in robotic perception tasks. With an operating range of 0.3-3 meters and HD resolution, it enables precise depth mapping for obstacle detection, gesture recognition, and SLAM. The USB-C interface ensures high-speed data transfer suitable for real-time applications. This system supports evaluation of depth-estimation algorithms and enhances spatial understanding in indoor and outdoor environments.

4. **MTi-G710 2A8G4DK (Xsens):** The MTi-G710 GNSS/INS module provides high-precision orientation, position, and motion data using a fusion of IMU, magnetometer, barometer, and multi-band GNSS. Designed for autonomous systems and robotics, it ensures reliable navigation even under challenging conditions such as multipath interference or partial signal loss. The module supports real-time attitude estimation and trajectory reconstruction, enabling accurate evaluation of motion-planning and localization algorithms.

5. **Xilinx PYNQ-Z2 Board:** The Xilinx PYNQ-Z2 is a Zynq-7000 SoC–based development board used for rapid prototyping of FPGA-accelerated algorithms. Its Python-based PYNQ environment allows seamless integration of programmable logic for hardware acceleration in image processing, machine learning, and control tasks. The platform enables experimentation with custom hardware overlays and supports real-time embedded system development.

6. **Xilinx ZCU104 Evaluation Kit:** The ZCU104 evaluation kit, powered by the Zynq UltraScale+ MPSoC, supports high-performance embedded processing and hardware acceleration for AI, DSP and Real-time Control applications. The board provides extensive I/O, PL resources, and multi-core ARM processing, enabling system-level development of perception pipelines, neural network inference engines, and heterogeneous computing architectures. It is ideal for validating compute-intensive algorithms in autonomous and edge-AI research.

7. **Buggy Vehicle:** A research-grade buggy vehicle serves as a mobile platform for autonomous navigation and perception experiments. It supports integration of LiDARs, Cameras, Radars and Embedded computers for real-world testing of control algorithms, sensor fusion, and path-planning. The platform's adaptability allows safe and repeatable evaluation of autonomous driving functions under controlled conditions.

8. **Logitech G29 Racing Wheel:** The Logitech G29 racing wheel is employed as a human-machine interface device for driver-behavior modeling, driving simulators, and control-system testing. Its force-feedback mechanism replicates realistic steering dynamics, enabling accurate studies of manual control inputs, driver reaction analysis, and semi-autonomous vehicle supervision tasks.

9. **Logitech G Driving Shifter:** The Logitech G driving shifter provides manual gear-shifting input for driving simulators and human-in-the-loop testing setups. Its tactile feedback and mechanical design support research on driver workload, interaction modeling, and gearbox-control strategies in vehicular simulation environments.

10. **PNY T1000 4GB Graphics Card:** The PNY T1000 GPU offers professional-grade graphics and compute acceleration for machine vision, simulation, and AI model development. With CUDA support and efficient power consumption, it facilitates medium-scale neural network training, real-time image rendering, and GPU-accelerated perception algorithm implementation in research workstations.

11. **AWR Radar:** An AWR-series mmWave radar unit is employed for high-frequency sensing tasks including range detection, speed estimation, and environmental perception. Its FMCW architecture enables robust operation in adverse conditions such as fog, darkness, or dust. The system supports advanced radar-signal-processing research for automotive and robotic applications.

12. **Q-CAR SDRS Package:** The Q-CAR SDRS package is an advanced autonomous vehicle research platform integrating sensors, compute units, and drive-by-wire systems. It allows full-stack testing of perception, planning, and control algorithms. The platform provides repeatable real-world evaluation scenarios, enabling development of ADAS and autonomous driving features.

13. **dSPACE ADAS-HIL Systems:** The dSPACE ADAS-HIL system provides real-time hardware-in-the-loop simulation for validating autonomous driving components. It enables virtual testing of perception, control, and communication modules under diverse traffic and environmental conditions. High-fidelity plant models and sensor simulations allow safe and extensive evaluation before real-world deployment.

14. **Workstations (HP Z6 G4):** HP Z6 G4 workstations serve as high-performance computing nodes for AI training, simulation, and data processing. Equipped with multi-core CPUs, large RAM capacity, and GPU expandability, they support parallel workloads such as deep learning inference, sensor data analysis, and embedded system toolchains. These workstations ensure reliable and efficient execution of compute-intensive research tasks.

15. **NVIDIA Xavier AGX Developer Kit:** The NVIDIA AGX Xavier developer kit provides an AI-optimized embedded computing platform supporting real-time perception and autonomous-robotics applications. With integrated GPU, NVDLA accelerators, and high-bandwidth memory, it enables deployment of advanced neural networks for vision, SLAM, and sensor fusion.

16. **Event Camera Sensor PSF138MPEV280:** The PSF138MPEV280 event-based sensor captures asynchronous pixel-level changes, enabling high-speed vision tasks such as motion tracking, neuromorphic processing, and low-latency perception. Its sparse data output supports energy-efficient computation and excels in high-dynamic-range environments. This sensor is well-suited for studying bio-inspired vision algorithms and rapid-motion robotics.

17. **32-bit ARM Cortex-M7 Evaluation Board:** A Cortex-M7–based development board is used for evaluating high-performance microcontroller applications in control, signal processing, and embedded robotics. Its DSP instructions and real-time processing capabilities enable implementation of fast control loops, sensor interfaces, and low-power edge-computing algorithms. The board provides a platform for firmware development and hardware-software co-design.

18. **USRP B210:**The USRP B210 software-defined radio platform is utilized for wideband wireless communication research and signal-processing experimentation. Covering frequencies from HF to 6 GHz with a full-duplex MIMO architecture, it enables real-time implementation of modulation schemes, spectrum sensing, and PHY-layer prototyping. The high-speed USB 3.0 interface supports streaming of complex baseband samples for cognitive radio development, 5G waveform testing, and RF machine-learning studies. Its flexible FPGA-based architecture allows rapid development and validation of custom communication algorithms in dynamic and controlled RF environments.

19. **ChipWhisperer Lite 2-Port with XMEGA:** The Chip-Whisperer Lite 2-port platform with XMEGA microcontroller is used for side-channel analysis (SCA) and hardware-security experimentation. Designed for power-analysis, fault-injection, and cryptographic vulnerability research, it enables high-precision capture of power traces during embedded firmware execution. The XMEGA target allows controlled execution of cryptographic algorithms such as AES for evaluating resistance against timing attacks, differential power analysis, and glitch-based fault attacks. This setup provides a comprehensive environment for studying embedded system security, validating countermeasures, and training machine-learning models for SCA-based threat detection.

20. **Skateboard:**The Autonomous Vehicle Skateboard is a modular robotic platform used for research in autonomous navigation and smart mobility systems. It integrates sensors, controllers, motor drivers, and battery systems for experimentation in obstacle detection, path planning, and autonomous control. The platform supports UG projects, research activities, and innovation in robotics and intelligent transportation technologies.

21. **Autonomous Buggy:** The Autonomous Buggy is a self-driving robotic vehicle platform equipped with sensors, embedded controllers and intelligent navigation algorithms. It is used for hands-on learning and research in autonomous driving, robotics, computer vision, path planning and artificial intelligence. The facility enables students to design, develop and test autonomous navigation and obstacle avoidance systems, supoorting project based learning and innovation in smart mobility and intelligent transportation applications.

PART E: First Year faculty and financial Resources

(Data to be filled in for the first year course faculty and budget allocation and utilization)

E1. First Year Student-Faculty Ratio (FYSFR)

Table No. E1.1: FYSFR details.

Year	Sanctioned intake of all UG programs (S4)	No. of required faculty (RF4= S4/20)	No. of faculty members in Basic Science Courses & Humanities and Social Sciences including Management courses (NS1)	No. of faculty members in Engineering Science Courses (NS2)	Percentage= No. of faculty members ((NS1*0.8) + (NS2*0.2))/(No. of required faculty (RF4)); Percentage= ((NS1*0.8) +(NS2*0.2))/RF
2023-24(CAYm2)	1380	69	47	20	60
2024-25(CAYm1)	1380	69	51	24	66
2025-26(CAY)	1800	90	59	25	58

E2. Budget Allocation, Utilization, and Public Accounting at Institute Level

Table No. E2.1: Budget and actual expenditure incurred at Institute level.

Items	Budgeted in 2025-26	Actual Expenses in 2025-26 till	Budgeted in 2024-25	Actual Expenses in 2024-25 till	Budgeted in 2023-24	Actual Expenses in 2023-24 till	Budgeted in 2022-23	Actual Expenses in 2022-23 till
Infrastructure Built-Up	275000000.00	266400332.26	292200000.00	337836058.50	320300000.00	275490296.08	226000000	272900679
Library	10800000.00	11640291.79	11600000.00	9771428.06	8100000.00	6934858.90	6550000	6684290
Laboratory equipment	77000000.00	70298745.00	40000000.00	42923569.00	30000000.00	42637137.00	60000000	45115063
Teaching and non-teaching staff salary	583700000.00	573446030.00	561500000.00	551937567.00	512840000.00	498123229.00	486223533	486223533
Outreach Programs	2000000.00	1821446.00	1700000.00	1059445.00	1200000.00	1185057.00	1759579.55	1759579.55
R&D	95000000.00	89863171.00	120000000.00	105715282.20	91000000.00	67685714.98	85916358	85916358

Training, Placement and Industry linkage	40000000.00	32995616.20	28000000.00	28096690.28	39400000.00	27868861.85	16700000	22804772.45
SDGs	500000.00	00	500000.00	000	00	00	00	00
Entrepreneurship	6000000.00	5500402.00	6500000.00	5765247.00	8500000.00	8288555.00	6000000	5721313
Others, specify	318247000.00	300391085.97	240450000.00	236975209.38	221575000.00	222576706.76	200300000	212374205
Total	**1408247000.00**	**1352357120.22**	**1302450000.00**	**1320080496.42**	**1232915000.00**	**1150790416.57**	**1089449470.55**	**1139499793.00**

E3. Budget Allocation, Utilization, and Public Accounting at Program Specific Level

Table No. E3.1: Budget and actual expenditure incurred at program level.

Items	**Budgeted in 2025-26**	**Actual Expenses in 2025-26 till**	**Budgeted in 2024-25**	**Actual Expenses in 2024-25 till**	**Budgeted in 2023-24**	**Actual Expenses in 2023-24 till**	**Budgeted in 2022-23**	**Actual Expenses in 2022-23 till**
Laboratory equipment	5000000.00	5435038.00	6300000.00	6159233.00	9100000.00	9027775.00	9000000.00	7753126.00
Software	1250000.00	540888.00	00	00	00	00	2000000.00	1770000.00
SDGs	50000.00	00	00	00	00	00	00	00
Support for faculty development	1800000.00	1604719.25	2200000.00	1466796.00	3500000.00	3158555.00	575000.00	564136.00
R & D	21700000.00	19880475.00	5600000.00	5401997.00	3900000.00	3949740.00	28980000.00	28888421.00
Industrial Training, Industry expert, Internship	7000000.00	6293398.00	5200000.00	5433727.00	3500000.00	2970561.00	2525000.00	5502715.00
Miscellaneous Expenses*	244050000.00	238764806.73	311750000.00	332469641.51	287452000.00	262304150.89	214940000.00	231479533.00
Total	**280850000.00**	**272519324.98**	**331050000.00**	**350931394.51**	**307452000.00**	**281410781.89**	**258020000.00**	**275957931.00**